

```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4  USE work.ALL;
5
6  -- Producing a synchronous edge-detect pulse for the leading and trailing
7  -- edges of a long (many clocks) asynchronous pulse.
8
9  ENTITY synch IS
10   PORT (
11     async_sig : IN STD_LOGIC;
12     clk       : IN STD_LOGIC;
13     rise      : OUT STD_LOGIC;
14     fall      : OUT STD_LOGIC);
15  END;
16
17  ARCHITECTURE RTL OF synch IS
18  BEGIN
19     sync1 : PROCESS (clk)
20       VARIABLE resync : STD_LOGIC_VECTOR(1 TO 3);
21     BEGIN
22       IF rising_edge(clk) THEN
23         -- detect rising and falling edges.
24         rise <= resync(2) AND NOT resync(3);
25         fall <= resync(3) AND NOT resync(2);
26         -- update history shifter.
27         resync := async_sig & resync(1 TO 2);
28       END IF;
29     END PROCESS;
30
31  END ARCHITECTURE;
```